

TEST SPECIFICATION DOCUMENT

Service Mobility Dashboard

Version 1 - August 2026

Document Type	Test Specification
Product	Service Mobility Dashboard
Status	Working baseline - to be refined before final submission

1. Purpose

The purpose of this document is to define the tests required to verify that the Service Mobility Dashboard operates correctly, reliably, and securely.

The testing process focuses primarily on the system's core business flows:

- Secure access to the dashboard.
- Loading service tasks for a selected workday.
- Displaying service tasks geographically.
- Grouping tasks by service vehicle.
- Calculating and displaying optimized routes.
- Viewing detailed service-task information.
- Searching for a new service address.
- Receiving vehicle-and-date recommendations.
- Switching between Hebrew and English.
- Preventing unauthorized modification of SAP-mirrored data.
- Handling failures and edge cases without making the dashboard unusable.

The testing strategy focuses on validating the product's core business flows, business rules, and critical system behavior rather than achieving exhaustive line-by-line code coverage.

2. Testing Strategy

Unit Tests

Unit tests will be used for deterministic business and utility logic that can be tested independently.

Examples include:

- Date validation and calendar calculations.
- Coordinate validation.
- Grouping tasks by vehicle.
- Map bounds calculations.
- Route coordinate formatting.
- Distance and duration formatting.
- Recommendation filtering and ordering.
- Safe redirect validation.

Component Tests

Component tests will verify important UI behavior in isolation.

Examples include:

- Login form behavior.
- Date picker interaction.
- Address search behavior.
- Task tooltip rendering.
- Language switching.

Integration Tests

Integration tests will verify interactions between multiple layers of the application.

Examples include:

- Server Actions communicating with Supabase.
- Authentication checks before returning operational information.
- Recommendation generation from database records.
- Mapbox API responses being converted into route data.

End-to-End Tests

End-to-End tests will verify the most important flows from the dispatcher's perspective.

Examples include:

- Login -> Dashboard -> Select Date -> View Tasks.
- Search Address -> Receive Recommendations -> Open Recommended Workday.
- Logout -> Return to Login page.

Manual Tests

Some map and visual behavior is more appropriately verified manually.

Examples include:

- Correct visual separation between vehicle routes.
- Map interaction and usability.
- Hebrew RTL presentation.
- Responsive layout.
- Visual fallback when a route cannot be calculated.

3. Test Environment

Tests should be performed against a controlled test environment containing known service-task records.

External services such as Mapbox should be mocked in automated tests where possible so that tests do not depend on network availability or external API rate limits.

A smaller set of manual or End-to-End tests may also be performed using the real Mapbox service.

The database test data should include:

- Multiple vehicles.
- Multiple tasks for the same vehicle.
- Tasks on several different dates.
- A vehicle with only one task.
- A vehicle with more than 12 tasks.
- Tasks with invalid coordinates.
- Tasks with optional information missing.
- Tasks located both inside and outside the 20 km recommendation radius.
- Multiple tasks belonging to the same vehicle on the same day.
- A date containing no service tasks.

4. Authentication and Authorization Tests

AUTH-01 - Unauthenticated User Attempts to Access Dashboard

Scenario: A user who is not logged in navigates directly to the dashboard.

Expected Result:

- The user is redirected to /login.
- Operational service-task information is not displayed.

AUTH-02 - Preserve Requested Date During Authentication

Scenario: An unauthenticated user opens a dashboard URL for a specific date.

`/?date=2026-08-20`

Expected Result:

- The user is redirected to the Login page.
- The original requested URL is preserved.
- After successful login, the user returns to the requested date rather than today's dashboard.

AUTH-03 - Successful Login

Scenario: An authorized user enters valid credentials.

Expected Result:

- Authentication succeeds.
- The user is redirected to the dashboard.
- The dashboard becomes accessible.

AUTH-04 - Invalid Credentials

Scenario: The user enters an invalid email or password.

Expected Result:

- Login fails.
- The dashboard remains inaccessible.
- The user receives the generic message: "Invalid email or password."
- The system does not reveal whether the email address itself exists.

AUTH-05 - Empty Login Fields

Scenario: The user attempts to submit the login form without an email or password.

Expected Result:

- Login is not performed.
- Required-field validation prevents the request or the server rejects it.
- No authenticated session is created.

AUTH-06 - Open Redirect Protection

Scenario: A malicious login URL attempts to use an external redirect destination.

`/login?redirectTo=https://malicious.example`

Expected Result:

- The system must not redirect the authenticated user to an external domain.

- The destination falls back to /.

AUTH-07 - Logout

Scenario: An authenticated dispatcher selects Sign Out.

Expected Result:

- The Supabase session is terminated.
- The user is redirected to /login.
- Returning to a protected dashboard URL requires authentication again.

5. Daily Task and Date Tests

DATE-01 - Default Date

Scenario: An authenticated dispatcher opens the dashboard without a date query parameter.

Expected Result:

- The dashboard displays today's date.
- "Today" is calculated according to the Asia/Jerusalem service timezone.
- Only tasks scheduled for that date are retrieved.

DATE-02 - Select Valid Date

Scenario: The dispatcher selects another workday.

Expected Result:

- The URL changes to /?date=YYYY-MM-DD.
- The server loads tasks for the selected date.
- Tasks from other dates are not displayed.

DATE-03 - Invalid Date in URL

Scenario: The dispatcher or an external link supplies an invalid or malformed date value.

/?date=2026-02-30

Expected Result:

- The invalid value is not sent as a database date query.
- The application falls back to today's date.
- The page remains usable.

DATE-04 - Previous and Next Day Navigation

Scenario: The dispatcher uses the previous-day or next-day controls.

Expected Result:

- The selected date changes by exactly one calendar day.
- The dashboard loads the corresponding tasks.

DATE-05 - Calendar Selection

Scenario: The dispatcher opens the calendar and selects a date.

Expected Result:

- The calendar closes.

- The selected date appears in the header.
- The dashboard loads the selected day's data.

DATE-06 - Date With No Tasks

Scenario: The dispatcher selects a date for which there are no service tasks.

Expected Result:

- The application does not crash.
- An appropriate empty state is displayed.
- The map remains usable.

6. Database and Task Retrieval Tests

DATA-01 - Retrieve Only Selected Day's Tasks

Scenario: The database contains tasks across multiple dates.

Expected Result:

- `getDailyTasks()` returns only tasks whose `scheduled_date` matches the requested date.

DATA-02 - Unauthenticated Server Action Request

Scenario: `getDailyTasks()` is invoked without a valid authenticated session.

Expected Result:

- No operational tasks are returned.
- The result contains an authentication error.

DATA-03 - Database Failure

Scenario: Supabase fails to return the service-task data.

Expected Result:

- The application does not crash.
- An empty task list is returned to the dashboard.
- An appropriate error message is displayed to the dispatcher.

7. Map and Vehicle Grouping Tests

MAP-01 - Group Tasks by Vehicle

Scenario: The selected day contains tasks assigned to multiple vehicles.

Expected Result:

- Tasks with the same `vehicle_id` are grouped together.
- Each vehicle is represented separately on the dashboard.
- Vehicle ordering remains deterministic between renders.

MAP-02 - Valid Coordinates

Scenario: A task contains valid latitude and longitude values.

Expected Result:

- The task is included in the geographical map data.

- It may participate in route optimization.

MAP-03 - Invalid Coordinates

Scenario: A task contains invalid geographical coordinates, such as latitude greater than 90, longitude greater than 180, or non-numeric coordinates.

Expected Result:

- The invalid task is not sent to the Mapbox Optimization API.
- It is excluded from geographical calculations.
- Other valid tasks continue to work normally.

MAP-04 - Map Bounds

Scenario: Several valid tasks exist at different locations.

Expected Result:

- The map automatically adjusts its visible bounds so that all valid service locations are visible.

8. Route Optimization Tests

ROUTE-01 - Vehicle With Multiple Stops

Scenario: A vehicle has at least two valid service-task coordinates.

Expected Result:

- A Mapbox Optimization request is created.
- The optimized route is displayed on the map.
- Stop numbers correspond to the order returned by Mapbox.

ROUTE-02 - Vehicle With One Stop

Scenario: A vehicle contains only one valid service task.

Expected Result:

- The task marker is displayed.
- No unnecessary Optimization API request is sent.
- No route line is required.

ROUTE-03 - More Than 12 Stops

Scenario: A vehicle has more than 12 tasks with valid coordinates.

Expected Result:

- At most 12 coordinates are sent to a single Mapbox Optimization API request.
- Remaining tasks continue to appear as markers.
- The application does not send an invalid Mapbox request.

ROUTE-04 - Successful Mapbox Response

Scenario: Mapbox returns a valid optimized trip.

Expected Result:

- The route geometry is displayed.
- Distance and duration values are stored for the route.

- The optimized waypoint order is mapped correctly to task IDs.

ROUTE-05 - Mapbox Optimization Failure

Scenario: The Optimization API fails, returns NoRoute, or cannot be reached.

Expected Result:

- Service-task markers remain visible.
- The entire dashboard does not fail.
- The dispatcher receives an appropriate error notification.
- Other vehicles whose optimization succeeded remain unaffected.

ROUTE-06 - Malformed Mapbox Response

Scenario: Mapbox returns a response without usable route geometry.

Expected Result:

- The response is treated as a route failure.
- An invalid route is not drawn.
- The remaining dashboard functionality continues to operate.

9. Service Task Detail Tests

TASK-01 - Open Task Details

Scenario: The dispatcher selects a task marker.

Expected Result:

- A task information popup appears.
- Available information includes relevant fields such as installer, vehicle plate number, customer name, customer phone number, address, service time window, operational note, and route stop number when available.

TASK-02 - Missing Optional Fields

Scenario: A service task is missing optional SAP information such as customer phone number or operational note.

Expected Result:

- The application does not display null or undefined.
- Empty detail rows are omitted.
- The rest of the task information remains readable.

10. Address Search Tests

SEARCH-01 - Short Search Input

Scenario: The dispatcher enters fewer than three characters.

Expected Result:

- No Geocoding API request is made.
- No suggestions are displayed.

SEARCH-02 - Valid Address Search

Scenario: The dispatcher types a valid Israeli address.

Expected Result:

- The system requests address suggestions from Mapbox Geocoding.
- Valid suggestions are displayed.
- Selecting a suggestion returns its latitude, longitude, and formatted address.

SEARCH-03 - Select Address

Scenario: The dispatcher selects one of the address suggestions.

Expected Result:

- The selected address remains visible in the search field.
- The suggestions dropdown closes.
- A marker for the selected location is displayed on the map.
- The map focuses on the selected location.

SEARCH-04 - Geocoding Failure

Scenario: The Mapbox Geocoding service fails or cannot be reached.

Expected Result:

- The application does not crash.
- Invalid or stale suggestions are not displayed.
- The user can continue using the existing dashboard.

11. Recommendation Tests

REC-01 - Four-Day Recommendation Window

Scenario: A new address is selected.

Expected Result:

- The system examines service tasks scheduled from tomorrow through the following four-day recommendation window.
- The recommendation window is based on the real current date rather than the date currently displayed on the dashboard.

REC-02 - Recommendation Radius

Scenario: The database contains tasks both inside and outside a 20 km radius of the selected address.

Expected Result:

- Only candidates at a distance of 20 km or less are returned.
- Tasks outside the radius are excluded.

REC-03 - Closest Task Per Vehicle and Date

Scenario: The same vehicle has multiple tasks on the same date within the recommendation radius.

Expected Result:

- Only the closest task for that vehicle/date combination appears as a recommendation.
- Duplicate vehicle/date recommendations are not displayed.

REC-04 - Recommendation Sorting

Scenario: Several valid recommendation candidates exist.

Expected Result:

- Recommendations are ordered from shortest geographical distance to longest geographical distance.

REC-05 - Invalid Task Coordinates

Scenario: Potential recommendation tasks contain invalid coordinates.

Expected Result:

- These tasks are excluded from the recommendation calculation.
- Valid recommendation candidates continue to be processed.

REC-06 - No Recommendations

Scenario: No upcoming service task is located within 20 km of the selected address.

Expected Result:

- The result contains no recommendation candidates.
- The dashboard remains usable.
- No false recommendation is generated.

REC-07 - Select Recommendation

Scenario: The dispatcher selects a recommended vehicle/date candidate.

Expected Result:

- The dashboard navigates to the candidate's `scheduled_date`.
- The searched address remains available for comparison.
- The dispatcher can inspect that day's existing work before making a scheduling decision.

REC-08 - Database Failure During Recommendation Search

Scenario: Supabase fails while recommendation candidates are being retrieved.

Expected Result:

- No incorrect recommendations are displayed.
- The rest of the dashboard remains operational.

12. Input Validation Tests

VAL-01 - Invalid Search-Date Input

Invalid date strings must not be sent directly to the database. Expected behavior is covered by DATE-03.

VAL-02 - Invalid Recommendation Coordinates

Scenario: The recommendation Server Action is invoked with coordinates outside the valid latitude/longitude ranges or with non-finite values.

Expected Result:

- The values should be rejected before performing recommendation calculations.
- No database-derived recommendation should be returned for invalid geographical input.

VAL-03 - External Redirect Input

External or protocol-relative redirectTo values must be rejected. Expected behavior is covered by AUTH-06.

13. Database Permission Tests**DB-01 - Authorized Read Access**

Scenario: An authorized application user retrieves service-task information.

Expected Result:

- The user can read the data required by the dashboard.

DB-02 - Prevent INSERT

Scenario: A normal authenticated or anon application role attempts to insert a service task.

Expected Result:

- PostgreSQL/Supabase rejects the operation.

DB-03 - Prevent UPDATE

Scenario: A normal application role attempts to modify an existing service task.

Expected Result:

- The database rejects the update.

DB-04 - Prevent DELETE

Scenario: A normal application role attempts to delete a service task.

Expected Result:

- The database rejects the deletion.

These tests verify the architectural requirement that the service_tasks table remains a read-only mirror of SAP.

14. Language and UI Tests**UI-01 - English Interface**

Scenario: English is selected.

Expected Result:

- Application text is displayed in English.
- The interface uses left-to-right direction where appropriate.

UI-02 - Hebrew Interface

Scenario: Hebrew is selected.

Expected Result:

- Application text is displayed in Hebrew.
- The main interface switches to right-to-left layout.
- The map remains fully functional.

UI-03 - Numeric Information in RTL Mode

Scenario: A Hebrew interface displays values such as phone numbers, vehicle plates, and time windows.

Expected Result:

- Values that are conventionally read left-to-right remain correctly ordered.
- Digits are not visually reversed by the RTL layout.

UI-04 - Loading States

Scenario: The application is waiting for route or recommendation information.

Expected Result:

- The interface provides visual feedback that processing is taking place.
- Repeated actions that would create conflicting requests are avoided where appropriate.

UI-05 - Error Messages

Scenario: A recoverable data or routing error occurs.

Expected Result:

- The user receives a visible and understandable error message.
- The error does not unnecessarily block unrelated functionality.

15. Important Edge Cases

The following edge cases must also be verified:

- A workday with zero service tasks.
- A vehicle with exactly one task.
- A vehicle with exactly 12 routable tasks.
- A vehicle with more than 12 routable tasks.
- Several vehicles with tasks in the same geographical area.
- Invalid coordinates in one task while all other tasks are valid.
- Missing optional customer information.
- External Mapbox failure for only one vehicle.
- Rapid switching between dates.
- Rapidly changing the searched address while a previous request is still running.
- Switching language while the dashboard is open.
- An expired authentication session.

16. Core End-to-End Business Flows

E2E-01 - Daily Planning Flow

1. Dispatcher opens the application.
2. Dispatcher signs in.

3. Dashboard loads the selected workday.
4. Service vehicles and tasks are displayed.
5. Optimized routes are calculated.
6. Dispatcher selects a task.
7. Task information is displayed.

Success Criteria: The complete daily operational view can be reached and used without an error.

E2E-02 - New Service Task Decision-Support Flow

8. Dispatcher signs in.
9. Dispatcher searches for a new service address.
10. Dispatcher selects the correct address.
11. The location appears on the map.
12. The system calculates upcoming vehicle/date recommendations.
13. Recommendations are displayed in distance order.
14. Dispatcher selects a recommendation.
15. Dashboard navigates to the relevant workday.
16. Dispatcher compares the new location with the existing schedule.

Success Criteria: The dispatcher can use the dashboard to evaluate a potential new-task assignment without modifying SAP data.

E2E-03 - Secure Logout Flow

17. Dispatcher is authenticated.
18. Dispatcher selects Sign Out.
19. Session is terminated.
20. Dispatcher is returned to the Login page.
21. Attempting to access the dashboard again redirects to Login.

Success Criteria: Operational information is inaccessible after logout.

17. Test Completion Criteria

The product will be considered ready for final submission when:

- All high-priority business-flow tests pass.
- Authentication and authorization tests pass.
- Read-only database protections are verified.
- Recommendation business rules are verified.
- Route failures are handled gracefully.
- No critical or high-severity defects remain open.
- Core functionality works in both Hebrew and English.
- Important edge cases have been tested.
- The production deployment has passed a final manual smoke test.